Exact Diffusion Design for Maximally Collective Stable Turing Patterns

in Binary-Complex Mass-Action Networks

Topology-Wide All-Spectrum Localization and Exponent-Optimal Stationary Heterogeneity Trade-Offs

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Abstract

Multicomponent Turing theories often seek a lower-dimensional unstable subsystem responsible for patterning. For every $n \geq 4$, we construct an n-species binary-complex classical mass-action topology for which, throughout every positive-equilibrium realization, every principal subsystem below order $n-1$ is Hurwitz while one order-$(n-1)$ subsystem is unstable. We prove a general principal-minor diffusion-ray theorem and obtain an exact network law: for any homogeneously stable realization, a stationary crossing occurs along a positive diagonal diffusion ray exactly when $d_Z > 8h_Z \sum_{j=2}^{n-2} d_j/h_j$. This gives the nonattained fixed-equilibrium contrast infimum, the unit-equilibrium value $8(n-3)$, and the topology-wide stationary trade-off $\chi_D\chi_H > 8(n-3)$. At unit equilibrium, an exact rational design undergoes a primary supercritical bifurcation with locally exponentially stable positive patterns in the fixed-integrated-mass class. A second exact family redistributes heterogeneity between diffusion and equilibrium concentrations so that both contrasts are $\Theta(\sqrt{n})$, matching the topology-specific stationary minimax lower bound in exponent. The proof uses explicit flux-cone design, exhaustive strongly connected block classification, conservation-compatible center-manifold reduction, and inspectable symbolic certificates. The networks are synthetic; the bifurcation, stability, and robustness conclusions are local for each fixed dimension, and the stationary diffusion law does not classify arbitrary wave instability.

Keywords: reaction–diffusion systems; Turing instability; mass-action kinetics; reaction networks; bifurcation; diffusion design.

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1 Introduction

Turing's diffusion-driven instability mechanism shows how a spatially homogeneous equilibrium can be stable in a well-mixed system and unstable to a nonconstant spatial mode after species-dependent diffusion is introduced [16]. For systems with many components, a central structural question is whether the instability can always be attributed to a smaller unstable subsystem. General matrix theories allow responsible principal subsystems of orders up to $n-1$ and distinguish stationary from wave instabilities [14, 1, 19, 18]. Random-matrix studies also find that diffusion thresholds often decrease with component count and that many sampled instabilities resist reduction to fewer diffusing species [5]. Reaction-network approaches instead constrain the Jacobian by stoichiometry, positive steady fluxes, and mass-action derivatives [11, 4, 15, 17]. Here "Turing bifurcation" means a stationary diffusion-driven crossing of a nonconstant Neumann mode on a fixed interval. We do not claim intrinsic finite-wavelength selection under enlargement of the spatial domain.

A large unstable block in one numerical Jacobian does not settle the reaction-topology question. Reparameterizing the same network can change its Jacobian and can expose a smaller unstable subsystem. Our first result closes this quantifier exactly: for one explicit indexed topology, every positive rate/equilibrium realization has all principal blocks below order $n-1$ Hurwitz. Thus the upper endpoint $n-1$ in principal-subsystem localization cannot be lowered even within a highly restricted classical mass-action class.

The second question is quantitative. Differential diffusion is often presented as an empirical tuning parameter, whereas the present topology admits a complete exact design law. A single signed order-$(n-1)$ minor is negative, and the remaining omission minors are positive or zero. This network-specific one-bad-minor structure reduces stationary diffusion design to one strict linear inequality and yields sharp heterogeneity lower bounds. We further show that the same topology supports primary supercritical bifurcations and locally exponentially stable patterns with substantially reduced diffusion contrast, as well as a stable diffusion–equilibrium trade-off whose two contrasts grow only as the square root of dimension.

Linear Turing instability alone does not guarantee a persistent nonlinear pattern [9]. We therefore prove the primary crossing, the conservation-compatible branch, the exact cubic sign, exchange of stability, and local semilinear stability separately. Related weakly nonlinear analyses for minimal two-species schemes demonstrate the importance of this additional layer [21, 22].

Theorem suite. For every $n = m + 1 \geq 4$, one indexed binary-complex topology has:

1. a complete family $A_m(a, b)H$ of all positive-equilibrium Jacobians, with two steady-flux parameters and arbitrary positive diagonal H;

2. topology-wide maximum all-spectrum localization order $n-1$;

3. an exact necessary-and-sufficient stationary diffusion criterion and a unique positive threshold on each eligible diffusion ray;

4. the sharp fixed-equilibrium diffusion-contrast infimum, including $8(m-2)$ at the unit equilibrium;
5. a rational unit-equilibrium supercritical stable-pattern realization;
6. a one-parameter stable family with $\chi_D, \chi_H = \Theta(\sqrt{m})$, matching the stationary minimax lower bound for this topology in exponent.

The topology was deliberately designed so that its positive-flux image and induced strongly connected blocks are explicit; no black-box real-algebraic elimination is used.

Scope. The network has one *semipositive* conservation law: the conserved functional does not contain X_1. Accordingly we call the system stoichiometric-codimension one, not conservative or mass conserving. The construction is synthetic and not proposed as a natural biochemical mechanism. All species diffuse with positive diagonal coefficients. The pattern, stability, and robustness statements are local for each fixed m; no arbitrary-data global existence, global attraction, dimension-uniform basin, cross-diffusion, or immobile-species theorem is claimed.

2 Reaction family and complete realization space

An indexed classical mass-action network has reactions $y_r \to y_r'$, source matrix $Y = [y_1 \ \cdots \ y_R]$, and stoichiometric matrix $\Gamma = [y_1' - y_1 \ \cdots \ y_R' - y_R]$. We call it *binary-complex* when

$$|y_r|_1 \le 2, \qquad |y_r'|_1 \le 2 \tag{1}$$

for every reaction. This condition restricts both source and product molecularity and is stronger than conventions that bound only reactant order.

At a positive equilibrium x^* define

$$v_r = k_r (x^*)^{y_r}, \qquad h_i = (x_i^*)^{-1}, \qquad H = \operatorname{diag}(h). \tag{2}$$

Then $v > 0$, $\Gamma v = 0$, and direct differentiation gives

$$J = \Gamma \operatorname{diag}(v) Y^T H. \tag{3}$$

Conversely, every $v > 0$ in $\ker \Gamma$ and every $H \succ 0$ are physically realized by $x_i^* = h_i^{-1}$ and $k_r = v_r/(x^*)^{y_r}$.

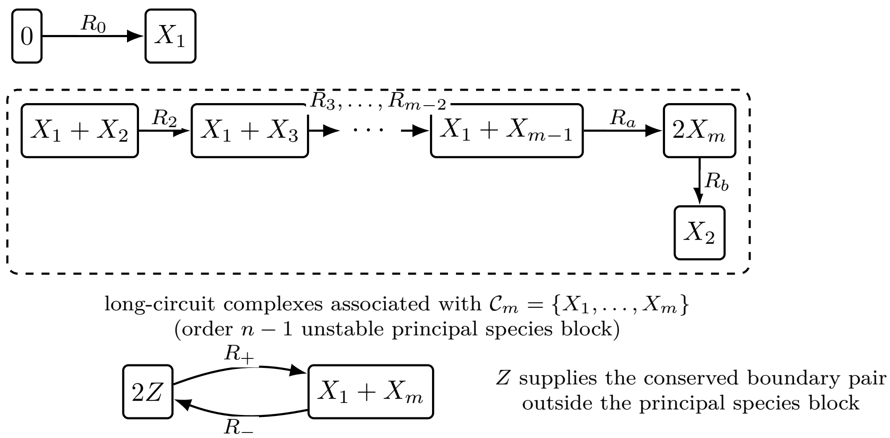
For $m \ge 3$, let $\widehat{N}_m$ have species $X_1, \ldots, X_m, Z$ and reactions

$$R_0: \quad 0 \longrightarrow X_1, \tag{4}$$

$$R_i: \quad X_1 + X_i \longrightarrow X_1 + X_{i+1}, \qquad 2 \le i \le m-2, \tag{5}$$

$$R_a: \quad X_1 + X_{m-1} \longrightarrow 2X_m, \tag{6}$$

$$R_b: \quad 2X_m \longrightarrow X_2, \tag{7}$$



Synthetic indexed topology; the semipositive conserved functional omits X_1.

Figure 1: The synthetic topology $\widehat{N}_m$. The dashed outline highlights the long-circuit complexes associated with the principal species block $\mathcal{C}_m = \{X_1, \dots, X_m\}$, the unique order-$(n-1)$ block with negative signed determinant. The separate Z boundary pair controls the exact diffusion law.

$$R_+ : \quad 2Z \longrightarrow X_1 + X_m, \tag{8}$$
$$R_- : \quad X_1 + X_m \longrightarrow 2Z. \tag{9}$$

The chain range is empty at $m = 3$. Writing $n = m + 1$, the family has n species and $n + 1$ indexed reactions and satisfies (1).

Proposition 2.1 (Semipositive conservation and flux cone)**.** *Let*

$$c = (0, \underbrace{4, \dots, 4}_{X_2, \dots, X_{m-1}}, 2, 1)^T. \tag{10}$$

Then

$$c^T \Gamma_m = 0, \qquad \operatorname{rank} \Gamma_m = m, \qquad \operatorname{im} \Gamma_m = c^\perp, \tag{11}$$

and

$$\ker \Gamma_m = \operatorname{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}. \tag{12}$$

Consequently

$$\{v > 0 : \Gamma_m v = 0\} = \{(a\mathbf{1}_m, b, b) : a, b > 0\}. \tag{13}$$

The functional $c^T x$ is semipositive: it does not bound X_1.

Proof. Each reaction vector is orthogonal to c. The X_i balance equations along the long circuit successively equate its m reaction fluxes, and the Z balance equates the final reversible pair. Thus these equations have exactly two independent free fluxes and the kernel is precisely the displayed two-dimensional space. Since Γ_m has $m + 2$ columns, rank–nullity gives $\operatorname{rank} \Gamma_m = m$. The image and positive-cone statements follow. $\square$

At unit equilibrium and flux $(a\mathbf{1}_m, b, b)$, the vector field is

$$\begin{aligned}
\dot{x}_1 &= a - ax_1x_{m-1} + bz^2 - bx_1x_m, \\
\dot{x}_2 &= -ax_1x_2 + ax_m^2, \\
\dot{x}_i &= ax_1x_{i-1} - ax_1x_i, && 3 \le i \le m-1, \\
\dot{x}_m &= 2ax_1x_{m-1} - 2ax_m^2 + bz^2 - bx_1x_m, \\
\dot{z} &= -2bz^2 + 2bx_1x_m.
\end{aligned} \tag{14}$$

The nonzero entries of its Jacobian factor $A_m(a, b)$ are

$$\begin{aligned}
&A_{11} = -(a+b), \quad A_{1,m-1} = -a, \quad A_{1m} = -b, \quad A_{1Z} = 2b, \\
&A_{21} = -a, \quad A_{22} = -a, \quad A_{2m} = 2a, \\
&A_{i,i-1} = a, \quad A_{ii} = -a, \qquad 3 \le i \le m-1, \\
&A_{m1} = 2a - b, \quad A_{m,m-1} = 2a, \quad A_{mm} = -(4a+b), \quad A_{mZ} = 2b, \\
&A_{Z1} = 2b, \quad A_{Zm} = 2b, \quad A_{ZZ} = -4b.
\end{aligned} \tag{15}$$

Proposition 2.2 (Complete realization family)**.** *Every positive-equilibrium Jacobian of $\widehat{N}_m$ is exactly*

$$J_m(a, b, H) = A_m(a, b)H, \qquad a, b > 0, \quad H \succ 0 \text{ diagonal.} \tag{16}$$

Conversely every triple in (16) *is realized by positive rates and a positive equilibrium.*

Proof. The forward direction is Eqs. (3) and (13). For the converse, choose $x_i^* = h_i^{-1}$, flux $(a\mathbf{1}_m, b, b)$, and $k_r = v_r/(x^*)^{y_r} > 0$. □

3 Maximum topology-wide all-spectrum localization

For a real square matrix M, write $\alpha(M) = \max\{\Re\lambda : \lambda \in \sigma(M)\}$.

Lemma 3.1 (Exhaustive strongly connected blocks)**.** *Let I be a principal species set with $0 < |I| < m$. Every strongly connected component of the induced Jacobian graph on I is either a singleton, whose corresponding diagonal entry is negative, or one of the following non-singleton blocks:*

1. *the complete long cycle $X_1, \dots, X_{m-1}$;*
2. *the complete long cycle $X_2, \dots, X_m$;*
3. *a non-singleton principal subgraph of the boundary triad $\{X_1, X_m, Z\}$.*

After a simultaneous permutation of rows and columns, every principal matrix with $0 < |I| < m$ is block upper triangular with diagonal blocks of these forms. On the hypersurface $b = 2a$, the edge $X_1 \to X_m$ is absent; this can only split components and hence refines the same classification.

Proof. Use the directed convention $X_j \to X_i$ when the (i, j) Jacobian entry is nonzero. For $m = 3$, inspect the principal sets of order at most two directly. The sets $\{X_1, X_2\}$ and $\{X_2, X_3\}$ are the two complete long cycles; every other non-singleton strongly connected component is a principal subgraph of $\{X_1, X_3, Z\}$. Hence assume $m \geq 4$ below. Along the interior chain the only off-diagonal arrows are $X_i \to X_{i+1}$ for $2 \leq i \leq m-2$. Thus a retained proper chain segment cannot close by itself.

A component containing an interior chain arrow can close in only two ways. If it reaches X_{m-1} and returns through $X_{m-1} \to X_1$, then strong connectivity forces every vertex $X_1, \ldots, X_{m-1}$: the only path from X_1 back through the chain successively uses $X_2, \ldots, X_{m-1}$. This is the first long cycle. If it reaches X_m and returns through $X_m \to X_2$, then strong connectivity forces every vertex $X_2, \ldots, X_m$; this is the second long cycle. The other potential chain closure $X_1 \to X_2 \to \cdots \to X_m \to X_1$ already uses m vertices and is excluded by $|I| < m$.

Each complete long cycle has $m - 1$ vertices. Adding any further boundary vertex therefore gives at least m retained vertices, again excluded. If no complete long cycle occurs, every feedback path involving no interior chain lies entirely inside $\{X_1, X_m, Z\}$: the arrows among those vertices are the only remaining off-diagonal boundary arrows. Every retained chain vertex outside such a feedback component has no return path and hence forms a singleton diagonal block with negative diagonal entry. Frobenius permutation gives the asserted block-triangular form. When $b = 2a$, the sole parameter-dependent arrow $X_1 \to X_m$ disappears. This arrow belongs to neither complete long cycle, so both long-cycle blocks are unchanged. Within the boundary graph, deleting an arrow can only split strongly connected components and cannot merge them, so no new case arises. □

Theorem 3.2 (Maximal all-spectrum localization)**.** *For every $m \geq 3$, $a, b > 0$, positive diagonal H, and principal set I with $0 < |I| < m$,*

$$\alpha(J_m(a, b, H)_{I,I}) < 0. \tag{17}$$

For $\mathcal{C}_m = \{X_1, \ldots, X_m\}$, the block $J_{\mathcal{C}_m,\mathcal{C}_m}$ has a positive real eigenvalue. Thus every positive realization of the topology satisfies

$$\boxed{\min\{|I| : \varnothing \neq I \subseteq [n], \alpha(J_{I,I}) > 0\} = m = n - 1.} \tag{18}$$

No smaller principal subsystem supports either stationary or oscillatory instability.

Proof. For the first long cycle, a closed-right-half-plane root would force

$$\prod_{i=1}^{m-1} |\lambda + \delta_i h_i| = a^{m-1} \prod_{i=1}^{m-1} h_i, \qquad \delta_1 = a + b, \quad \delta_i = a \ (i \geq 2),$$

but the left side is at least $(a + b)a^{m-2} \prod h_i > a^{m-1} \prod h_i$. For the second long cycle, the corresponding diagonal product is $(4a + b)a^{m-2} \prod h_i$, strictly larger than the cycle product $4a^{m-1} \prod h_i$.

The full boundary triad has characteristic polynomial $\lambda^3 + c_1\lambda^2 + c_2\lambda + c_3$, where

$$\begin{aligned} c_1 &= ah_1 + 4ah_m + bh_1 + bh_m + 4bh_Z, \\ c_2 &= a(4ah_1h_m + 7bh_1h_m + 4bh_1h_Z + 16bh_mh_Z), \\ c_3 &= 16a^2bh_1h_mh_Z. \end{aligned} \tag{19}$$

All coefficients are positive, and $c_1c_2 - c_3$ is a times fourteen strictly positive monomials (Supplement, Section S2). The cubic Routh–Hurwitz criterion and direct one- and two-dimensional checks establish Hurwitz stability of every triad principal block. The result for all smaller sets follows from Lemma 3.1 and Frobenius block triangularization.

Finally, the explicit lower-bidiagonal Schur-complement elimination in Supplement, Section S2, gives

$$(-1)^m \det J_{\mathcal{C}_m,\mathcal{C}_m} = -2a^{m-1}b\prod_{i=1}^{m} h_i < 0. \tag{20}$$

Its characteristic polynomial is negative at zero and positive for large positive λ, so the block has a positive real eigenvalue. □

Corollary 3.3 (Mass-action sharpness at the subsystem endpoint)**.** *General n-component matrix theory permits unstable (activator) subsystem dimensions through $p = n-1$ [14]. This endpoint is sharp even within binary-complex classical mass action: for every $n \geq 4$, one indexed topology has every principal subsystem below order $n-1$ Hurwitz throughout every positive realization, while an order-$(n-1)$ subsystem is unstable. Moreover, the topology admits a selected positive realization whose full reaction–diffusion system has locally stable patterns.*

Anma, Sakamoto, and Yoneda [1] provide complementary analysis of stable–unstable and unstable–unstable pairs of subsystems in the three-component setting, distinguishing steady and wave instability.

4 A principal-minor diffusion-ray theorem

Theorem 4.1 (Principal-minor diffusion ray)**.** *Let $n \geq 2$. For $I \subseteq [n]$, set $a_I = (-1)^{|I|}\det J_{I,I}$ and $a_\varnothing = 1$. Let $J \in \mathbb{R}^{n\times n}$ satisfy*

$$\det J = 0, \qquad a_I > 0 \quad (|I| \leq n-2), \qquad \sum_{|I|=n-1} a_I > 0. \tag{21}$$

For $D = \operatorname{diag}(d_1, \ldots, d_n) \succ 0$, define $p_D(s) = \det(sD - J)$. Then

$$p_D(s) = s\{\beta_1(D) + \beta_2(D)s + \cdots + \beta_n(D)s^{n-1}\}, \tag{22}$$

where

$$\beta_k(D) = \sum_{|I|=n-k} a_I \prod_{j\notin I} d_j, \qquad \beta_k(D) > 0 \quad (k \geq 2). \tag{23}$$

A nonzero stationary threshold exists on the ray sD if and only if $\beta_1(D) < 0$. In that case there is a unique simple $s_* > 0$, and

$$J - sD \text{ has a positive real eigenvalue} \quad \Longleftrightarrow \quad 0 < s < s_*. \tag{24}$$

For $s > s_*$ no positive real eigenvalue exists. The theorem does not exclude a nonreal unstable pair outside this interval.

Proof. Principal-minor expansion gives Eqs. (22) and (23); the constant term vanishes because $\det J = 0$. For $k \geq 2$, every summand has $|I| \leq n-2$ and is strictly positive. Hence the bracket in (22) is strictly increasing on $s \geq 0$, proving the threshold criterion, uniqueness, and scalar simplicity of the dispersion root s_*. Ordinary algebraic simplicity of the zero eigenvalue of $J - s_*D$ follows below from $\chi'_{s_*}(0) > 0$.

For fixed $s > 0$,

$$\chi_s(\lambda) = \det(\lambda I + sD - J) = \sum_I a_I \prod_{j \notin I} (\lambda + sd_j). \tag{25}$$

In $\chi'_s(\lambda)$, every contribution with $|I| \leq n-2$ is positive for $\lambda \geq 0$. The total contribution from $|I| = n-1$ is the positive number $\sum_{|I|=n-1} a_I$ in (21), while the full-order contribution vanishes. Therefore $\chi'_s(\lambda) > 0$ for all $\lambda \geq 0$. Since $\chi_s(0) = p_D(s)$ and $\chi_s(\lambda) \to \infty$, the exact band (24) follows. At $s = s_*$, $\chi_{s_*}(0) = 0$ and $\chi'_{s_*}(0) > 0$; hence zero is an algebraically simple eigenvalue of $J - s_*D$. □

Remark 4.2. The coefficient hypotheses (21) are the theorem's exact domain. They hold, in particular, when zero is algebraically simple, the remaining spectrum is stable, and all principal blocks of order at most $n-2$ are Hurwitz: the order-$(n-1)$ sum is the positive coefficient of λ in $\det(\lambda I - J)$. No restriction on the number of negative order-$(n-1)$ minors is needed in the general theorem.

5 Exact diffusion design and sharp linear contrast

The all-spectrum theorem supplies the lower-order hypotheses of Theorem 4.1. The remaining information is the complete order-m omission table.

Proposition 5.1 (Order-$(n-1)$ omission minors)**.** *For $J = A_m(a,b)H$,*

$$(-1)^m \det J_{\widehat{Z},\widehat{Z}} = -2a^{m-1}b \prod_{i=1}^{m} h_i, \tag{26}$$

$$(-1)^m \det J_{\widehat{X_j},\widehat{X_j}} = 16a^{m-1}bh_Z \prod_{\substack{1 \leq i \leq m \\ i \neq j}} h_i, \qquad 2 \leq j \leq m-1, \tag{27}$$

$$\det J_{\widehat{X_1},\widehat{X_1}} = \det J_{\widehat{X_m},\widehat{X_m}} = 0. \tag{28}$$

Consequently

$$\beta_1(D) = 2a^{m-1}b\left(\prod_{i=1}^{m} h_i\right)\left[8h_Z\sum_{j=2}^{m-1}\frac{d_j}{h_j} - d_Z\right]. \tag{29}$$

Proof. Positive right multiplication by H multiplies each principal determinant by the product of its selected h_i, so it suffices to identify the unit-H coefficients.

Omitting Z leaves the complete X-chain block. The Schur-complement calculation in Supplement S2 gives (20), hence (26). Now omit an interior species X_j, $2 \le j \le m-1$. A Frobenius permutation orders the right chain fragment $X_{j+1}, \dots, X_{m-1}$, then $\{X_1, X_m, Z\}$, then the left fragment $X_2, \dots, X_{j-1}$; empty fragments are omitted. The resulting matrix is block lower triangular. The $m-3$ retained interior chain vertices outside the boundary block contribute signed singleton factors ah_i. The boundary triad contributes

$$(-1)^3 \det J_{\{X_1,X_m,Z\},\{X_1,X_m,Z\}} = 16a^2b\,h_1h_mh_Z,$$

the constant coefficient in (19). Multiplication gives (27), including the factor a^{m-3}. If X_1 is omitted, the restriction of the semipositive conservation vector is a nonzero left nullvector. If X_m is omitted, the restriction of $H^{-1}(2,-2,\dots,-2,0,1)^T$ is a nonzero right nullvector because its X_m coordinate is zero. This proves (28).

In the coefficient β_1, each order-m signed minor is multiplied by the diffusivity of its omitted species. Factoring the common positive quantity yields (29). The structural coefficient 8 is exactly 16/2: the positive weight of each interior omission divided by the magnitude of the unique negative Z omission. □

Define the homogeneously stable realization domain

$$\mathfrak{S}_m = \{(a,b,H) : a,b > 0,\ H \succ 0 \text{ diagonal},\ J_m(a,b,H)|_{c^\perp} \text{ is Hurwitz}\}. \tag{30}$$

The complete region $\mathfrak{S}_m$ is not classified here. In particular, the necessary inequality $T(H) > 1$ below is not asserted to be sufficient for homogeneous stability.

Theorem 5.2 (Exact stationary diffusion law)**.** *Assume $(a,b,H) \in \mathfrak{S}_m$, and let the diffusion matrix be positive diagonal:*

$$D = \operatorname{diag}(d_1, \dots, d_m, d_Z) \succ 0.$$

A nonzero stationary crossing occurs on the positive ray sD if and only if

$$\boxed{d_Z > 8h_Z\sum_{j=2}^{m-1}\frac{d_j}{h_j}.} \tag{31}$$

When (31) *holds, the threshold $s_*(a,b,H,D)$ is unique and simple, and the exact positive-real-eigenvalue interval is $0 < s < s_*(a,b,H,D)$. Scaling $D \mapsto \gamma D$ gives*

$$s_*(a,b,H,\gamma D) = \frac{s_*(a,b,H,D)}{\gamma}.$$

Proof. Because $c^T J = 0$ and $J|_{c^\perp}$ is Hurwitz, the restriction to $c^\perp$ is invertible. Hence J has rank $n-1$, its conservation zero is algebraically simple, and the coefficient of λ in $\det(\lambda I - J)$ is positive:

$$\sum_{|I|=n-1} (-1)^{|I|} \det J_{I,I} > 0.$$

Thus all hypotheses of Theorem 4.1 hold, and combining it with Proposition 5.1 proves the claim. A second derivation uses the left vector c and the right zero vector

$$r_0 = H^{-1}(2, -2, \ldots, -2, 0, 1)^T.$$

Then

$$\lambda_0'(0) = -\frac{c^T D r_0}{c^T r_0}, \qquad c^T D r_0 = \frac{d_Z}{h_Z} - 8 \sum_{j=2}^{m-1} \frac{d_j}{h_j},$$

while

$$c^T r_0 = h_Z^{-1} - 8 \sum_{j=2}^{m-1} h_j^{-1} < 0.$$

The last inequality follows from the positive equal-damping coefficient $\beta_1(\mathbf{1}) > 0$. Hence the conservation eigenvalue initially moves right exactly under (31). □

For a Neumann interval $(0, \mathcal{L})$, mode k has $s = q_k^2 = (k\pi/\mathcal{L})^2$. Therefore every nonzero mode with $0 < q_k^2 < s_*(a, b, H, D)$ has a positive real eigenvalue, and no mode with $q_k^2 > s_*(a, b, H, D)$ has one. A separate spectral certificate is required to exclude wave instability for a chosen diffusion profile.

Theorem 5.3 (Sharp contrast and heterogeneity trade-off)**.** *Fix $(a, b, H) \in \mathfrak{S}_m$ and define*

$$T(H) = 8h_Z \sum_{j=2}^{m-1} \frac{1}{h_j}, \qquad \chi_D = \frac{d_{\max}}{d_{\min}}, \qquad \chi_H = \frac{h_{\max}}{h_{\min}} = \frac{x^*_{\max}}{x^*_{\min}}. \tag{32}$$

The quantity χ_H measures equilibrium concentration-scale separation. We do not interpret it as a universal measure of biological expense or implausibility. Homogeneous stability implies $T(H) > 1$, and

$$\inf\{\chi_D : D \succ 0 \text{ satisfies } (31)\} = T(H). \tag{33}$$

The infimum is not attained. At $H = I$,

$$\inf \chi_D = 8(m-2). \tag{34}$$

Every stationary crossing of this topology satisfies the strict topology-wide trade-off

$$\boxed{\chi_D \chi_H > 8(m-2).} \tag{35}$$

The product lower bound is sharp as an infimum over homogeneously stable realizations admitting a stationary crossing.

Proof. If (31) holds, then $d_Z > d_{\min}T(H)$ and $d_{\max} \geq d_Z$, so $\chi_D > T(H)$. Conversely set all X-species diffusivities to one and $d_Z = T(H) + \varepsilon$. This proves (33) and its strict nonattainment. The unit formula follows immediately. Since $h_Z/h_j \geq 1/\chi_H$, one has $T(H) \geq 8(m-2)/\chi_H$, yielding (35). The unit realization belongs to $\mathfrak{S}_m$; taking $H = I$ and the approaching diffusion sequence shows sharpness over homogeneously stable realizations admitting a stationary crossing. Equal diffusion cannot destabilize because $J - sdI$ shifts the stable nonzero spectrum and the conservation zero strictly left. □

6 A supercritical stable unit-equilibrium design

Set $a = b = 1$, $H = I$, $x^* = \mathbf{1}$, and

$$K_i = 91m - 181 - i. \tag{36}$$

Define the critical vector $r = (r_1, \ldots, r_m, r_Z)^T$ by

$$r_1 = 1, \quad r_i = -\frac{K_i}{63(m-2)} \ (2 \leq i \leq m-1), \quad r_m = -\frac{2}{9}, \quad r_Z = \frac{5}{14}, \tag{37}$$

and diffusion profile

$$d_1 = \frac{23}{63}, \quad d_i = \frac{1}{K_i} \ (2 \leq i \leq m-1), \quad d_m = \frac{1}{7}, \quad d_Z = \frac{16}{45}. \tag{38}$$

All entries are positive. Here $d_{\min} = d_2 = (91m - 183)^{-1}$ and $d_{\max} = d_1 = 23/63$, so

$$\chi_D^{\text{unit}}(m) = \frac{23(91m - 183)}{63}. \tag{39}$$

Let $\chi_{\text{stable}}(m)$ denote the infimum of diffusion contrast, at unit equilibrium, over positive realizations and positive diagonal diffusion profiles admitting a primary stationary bifurcation with a nearby positive, patterned branch that is locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space.

Theorem 6.1 (Primary supercritical stable design)**.** *On $(0, \pi)$ with Neumann conditions, consider*

$$\partial_t x = f_m(x) + (1-\mu) D_m \partial_{\xi\xi} x, \tag{40}$$

with (38)*. For every $m \geq 3$:*

1. *the well-mixed Jacobian, equivalently the homogeneous $k = 0$ mode restricted to $c^{\perp}$, is stable;*

2. *at $\mu = 0$ the first nonzero mode has one algebraically simple zero with right vector* (37)*; every other first-mode eigenvalue and every higher mode is stable, and no complex pair is critical;*

3. *the crossing is transverse;*

4. *in the adjoint-projection coordinate, the one-dimensional center-manifold normal form is*
$$\dot{A} = \eta_m \mu A + c_m A^3 + O(|A|^5 + |A|^3|\mu| + |A||\mu|^2), \qquad \eta_m > 0, \quad c_m < 0; \tag{41}$$
setting $\dot{A} = 0$ gives the associated stationary Lyapunov–Schmidt bifurcation equation;

5. *for sufficiently small $\mu > 0$, the two positive nonconstant branches are locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space.*

Consequently
$$8(m-2) \le \chi_{\text{stable}}(m) \le \frac{23(91m-183)}{63}. \tag{42}$$

Proof architecture. The homogeneous characteristic determinant is

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \qquad \lambda q_m = (1+\lambda)^{m-3} P(\lambda) - R(\lambda), \tag{43}$$

where $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. For $m = 3$, $q_3 = (\lambda+7)(\lambda^2+5\lambda+2)$ is Hurwitz. For $m \ge 4$, the exact difference $|1+\lambda|^2|P|^2 - |R|^2$ has 35 nonnegative monomials in $\Re\lambda$ and $(\Im\lambda)^2$, with equality only at the origin. Thus a nonzero closed-right-half-plane root of q_m would contradict $(1+\lambda)^{m-3}P = R$. Since $q_m(0) = 16m - 34 > 0$, the conservation zero is simple and the complementary homogeneous spectrum is stable.

For damping tD_m, sparse chain elimination gives

$$\det(\lambda I - A_m + tD_m) = Q_m F - G, \tag{44}$$

where $Q_m = \prod_{i=2}^{m-1}(\lambda + 1 + t/K_i)$ and the fixed boundary factors are

$$g_1 = \lambda + 2 + \frac{23}{63}t, \quad g_m = \lambda + 5 + \frac{1}{7}t, \quad g_Z = \lambda + 4 + \frac{16}{45}t,$$
$$F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z, \qquad G = g_Z(4g_1 + g_m) - 36.$$

At onset $\prod_{i=2}^{m-1}(1 + 1/K_i) = 91/90$. Retaining one exact chain factor yields a 77-term nonnegative modulus certificate, strict except at $(\lambda, t) = (0, 1)$. To determine the multiplicity at that equality point, put $\mathfrak{h}_m = \sum_{j=1}^{m-2} K_j^{-1}$ and

$$\Pi_m(\lambda) = \det(\lambda I - A_m + D_m) = Q_m(\lambda, 1)F(\lambda, 1) - G(\lambda, 1).$$

Exact differentiation gives

$$\Pi_m'(0) = \frac{7043400m - 13600927 - 7043400\mathfrak{h}_m}{255150} = -\frac{163}{45}\,\ell^T r > 0. \tag{45}$$

Hence $\lambda = 0$ is an algebraically simple eigenvalue of $A_m - D_m$; in particular, its kernel and cokernel are one-dimensional. Together with the 77-term exclusion, this proves the complete primary-mode statement. An exact left vector $\ell = (\ell_1, \ldots, \ell_m, \ell_Z)^T$ is

$$\ell_1 = -\frac{266}{815}, \quad \ell_i = \frac{78260(m-2)}{163(91m - 180 - i)}, \quad \ell_m = \frac{18368}{7335}, \quad \ell_Z = 1, \tag{46}$$

with $2 \le i \le m-1$. Exact harmonic-sum formulas give $\ell^T r < 0$ and $\ell^T D_m r < 0$, hence $\eta_m = (\ell^T D_m r)/(\ell^T r) > 0$.

The reaction field is quadratic: $f_m(\mathbf{1} + u) = A_m u + \frac{1}{2} B_m(u, u)$, with $c^T B_m = 0$. Since $\cos^2 \xi = (1 + \cos 2\xi)/2$, the stable corrections solve

$$A_m w_0 = -\frac{1}{4} B_m(r, r), \quad c^T w_0 = 0, \qquad (A_m - 4D_m) w_2 = -\frac{1}{4} B_m(r, r). \tag{47}$$

The interior w_2 recurrence telescopes through four consecutive K_i factors. Fourier projection gives

$$c_m = \frac{\ell^T \{B_m(r, w_0) + \frac{1}{2} B_m(r, w_2)\}}{\ell^T r}. \tag{48}$$

Its numerator reduces to $R_m + C_m \mathfrak{h}_m$. The exact shifted-polynomial certificate proves the numerator positive and the denominator negative for all $m \ge 3$, hence $c_m < 0$.

The sectorial fixed-mass semiflow has a one-dimensional center manifold [6, Chapter 6]. In the adjoint-projection coordinate its reduced vector field has the coefficients η_m, c_m above. The reflection $(\mathcal{R}u)(\xi) = u(\pi - \xi)$ commutes with the Neumann reaction–diffusion semiflow, preserves the fixed integrated-mass class, and sends the critical eigenfunction $r \cos \xi$ to $-r \cos \xi$. The center manifold may therefore be chosen reflection-equivariant, and in the adjoint-projection coordinate $\mathcal{R}$ acts as $A \mapsto -A$. Consequently the reduced vector field is odd in A; setting it to zero gives the associated stationary Lyapunov–Schmidt equation, and the two nonzero branches are exchanged by reflection. On the fixed-integrated-mass Neumann tangent spaces

$$X_c = \left\{ u \in H^2_N : \int_0^\pi c^T u \, d\xi = 0 \right\}, \qquad Y_c = \left\{ g \in L^2 : \int_0^\pi c^T g \, d\xi = 0 \right\},$$

the identity $c^T A_m = 0$ and the Neumann boundary conditions make the fixed-mass subspaces invariant. Let $\mathcal{L}_0 u = A_m u + D_m \partial_{\xi\xi} u$. The Neumann cosine expansion $u = \sum_{k \ge 0} u_k \cos(k\xi)$ reduces $\mathcal{L}_0$ to the matrix blocks

$$\mathcal{L}_{0,k} = A_m - k^2 D_m. \tag{49}$$

For $k = 0$, the fixed-mass condition is $c^T u_0 = 0$, and homogeneous stability makes $A_m : c^\perp \to c^\perp$ invertible. At $k = 1$, (45) gives a simple kernel $\operatorname{span}\{r\}$ and range $\ell^\perp$; every block with $k \ge 2$ is invertible by the all-mode certificate. Moreover, for all sufficiently large k,

$$A_m - k^2 D_m = -k^2 D_m \left(I - k^{-2} D_m^{-1} A_m \right), \qquad \left\| (A_m - k^2 D_m)^{-1} \right\| \le \frac{2 \| D_m^{-1} \|}{k^2}. \tag{50}$$

Together with the finitely many low-mode inverses, this k^{-2} bound maps the compatible L^2 Fourier data into H^2_N. Hence the range is closed and

$$\ker \mathcal{L}_0 = \operatorname{span}\{r\cos\xi\}, \qquad \operatorname{ran}\mathcal{L}_0 = \left\{ g \in Y_c : \int_0^\pi \ell^T g \cos\xi \, d\xi = 0 \right\}.$$

Thus $\mathcal{L}_0 : X_c \to Y_c$ is Fredholm of index zero, with cokernel $\operatorname{span}\{\ell\cos\xi\}$. The parameter derivative applied to the critical direction is $D_m r\cos\xi$, and its cokernel pairing is $(\pi/2)\ell^T D_m r \neq 0$. Thus the stationary simple-eigenvalue theorem of Crandall–Rabinowitz supplies the local bifurcating curve [3]; oddness and the signs above give its two halves,

$$A_\pm(\mu) = \pm\sqrt{-\eta_m\mu/c_m} + O(\mu^{3/2}).$$

Since the base equilibrium is $\mathbf{1}$ and, for each fixed m, the branch perturbation tends to zero in $H^2_N \hookrightarrow C^0$, both reflection-related equilibria remain componentwise strictly positive for sufficiently small $\mu > 0$. The center eigenvalue is $-2\eta_m\mu + O(\mu^2) < 0$; the complementary gap persists. Indeed, at onset

$$\alpha(A_m - k^2 D_m) \le \left\| \frac{A_m + A_m^T}{2} \right\|_2 - k^2 d_{\min} \longrightarrow -\infty,$$

so the all-mode certificate leaves a strictly positive complementary spectral gap after only finitely many modes are checked. Along the small branch the linearization has fixed domain H^2_N and is an analytic sectorial perturbation of positive diagonal Neumann diffusion: the reaction Jacobian is a bounded multiplication operator and the diffusion entries retain a positive lower bound. Standard elliptic coercivity and resolvent estimates therefore prevent spectrum from entering from infinity, while Kato's stability and analytic perturbation theories continue the finitely many isolated eigenvalues and their spectral projections [7, Chapters IV and VII]. The Neumann diffusion generator on the real fixed-mass L^2 space is sectorial with compact resolvent, and its half-order phase space is H^1; in one dimension the quadratic Nemytskii map is smooth from H^1 to L^2. Henry's principle of linearized stability then gives local exponential asymptotic stability in H^1 [6, Chapter 5]. For complementary local bifurcation and stability background, see [8, Chapter I]. The complete coefficient tables and algebra appear in the Supplement. □

7 A stable diffusion–equilibrium trade-off family

Put $\nu = m - 2$ and retain the effective diffusion matrix Δ_m from (38). Define

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \ge 2, \end{cases} \qquad L_0(\nu) = \frac{\kappa_\nu}{\sqrt{\nu}}, \qquad L_1(\nu) = \frac{90\nu}{90\nu + 1}. \tag{51}$$

For $\nu \ge 2$, the lower endpoint is the boundary of the homogeneous modulus estimate $\nu L^2 \ge 5/4$ used below. The exceptional case $\nu = 1$ is proved directly. Each sufficient endpoint is not asserted to be an intrinsic dynamical boundary of the family. For $L \in [L_0(\nu), L_1(\nu)]$, set

$$h_1 = h_m = h_Z = 1, \qquad h_i = \frac{K_i}{LK_{i-1}} \quad (2 \le i \le m-1), \qquad \mathsf{H}_m(L) = \mathrm{diag}(h), \tag{52}$$

and choose

$$D_m^{\mathrm{phys}}(L) = \mathsf{H}_m(L)\Delta_m, \qquad x^* = \mathsf{H}_m(L)^{-1}\mathbf{1}. \tag{53}$$

Positive rates realizing unit reaction flux are given by $k_r = (x^*)^{-y_r}$.

Theorem 7.1 (Stable exponent-optimal trade-off)**.** *On $(0, \pi)$ with homogeneous Neumann boundary conditions, consider the physical mass-action realization (52)–(53) governed by*

$$\partial_t x = f_m(\mathsf{H}_m(L)x) + (1-\mu)D_m^{\mathrm{phys}}(L)\partial_{\xi\xi}x.$$

For every $m \ge 3$ and $L \in [L_0(\nu), L_1(\nu)]$, at $\mu = 0$ this system has a primary simple transverse first-mode stationary bifurcation and a negative cubic coefficient. For all sufficiently small $\mu > 0$, it has two nearby positive patterned equilibria that are locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space. Its physical contrasts are

$$\chi_D(L) = \frac{23}{63}\, 91\nu L, \tag{54}$$

$$\chi_H(L) = \frac{91\nu - 1}{91\nu L}, \tag{55}$$

$$\chi_D(L)\chi_H(L) = \frac{23(91\nu - 1)}{63}. \tag{56}$$

This product equals the unit-equilibrium stable-profile contrast:

$$\chi_D(L)\chi_H(L) = \chi_D^{\mathrm{unit}}(m) = \frac{23(91m - 183)}{63}.$$

Thus L reallocates a fixed contrast product between physical diffusion and equilibrium concentration scales; it does not reduce the product. Throughout the certified interval, $\chi_D(L) > \chi_H(L)$, so $\max(\chi_D, \chi_H) = \chi_D(L)$ is strictly increasing in L and is uniquely minimized within this family at $L = L_0$. At $L = L_0$,

$$\chi_D = \frac{2093}{63}\kappa_\nu\sqrt{\nu}, \qquad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu}\frac{91\nu - 1}{91\nu}. \tag{57}$$

Thus both contrasts, and hence their maximum, are $\Theta(\sqrt{m})$ at the lower endpoint, compared with $\Theta(m)$ for the unit-equilibrium stable profile. Since every stationary crossing obeys

$$\max(\chi_D, \chi_H) > \sqrt{8(m-2)}, \tag{58}$$

the simultaneous exponent $1/2$ is optimal among stationary-crossing realizations of $\widehat{N}_m$. No constant-optimality or complete global Pareto-frontier claim is made.

Proof architecture. In normalized concentrations $\widehat{x} = \mathsf{H}_m(L)x$, the parameterized PDE is

$$\partial_t \widehat{x} = \mathsf{H}_m(L)\{f_m(\widehat{x}) + (1-\mu)\Delta_m \partial_{\xi\xi}\widehat{x}\}. \tag{59}$$

At $\mu = 0$, the stationary bracket $f_m(\widehat{x}) + \Delta_m \partial_{\xi\xi}\widehat{x}$ and its critical right vector are unchanged. The dynamic left vector is $\widetilde{\ell}(L) = \mathsf{H}_m(L)^{-1}\ell$, and the physical fixed-mass covector becomes $\mathsf{H}_m(L)^{-1}c$.

For $2 \le i \le m-1$, chain elimination gives

$$g_i = \frac{\lambda}{h_i} + 1 + \frac{t}{K_i} = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

If $t \ge 1$ and $\lambda = x + iy$ with $x \ge 0$, then

$$|Q_m|^2 \ge \left(\frac{91}{90}\right)^2 |1 + L\lambda|^{2\nu} \ge \left(\frac{91}{90}\right)^2 \left(1 + 2\nu Lx + \frac{1}{3}y^2\right),$$

because $\nu L^2 \ge 1/3$ throughout the new interval. The corresponding boundary difference has 84 nonnegative monomials and equality only at $(\lambda, t) = (0, 1)$.

The homogeneous mode requires a different boundary polynomial. At $t = 0$,

$$\frac{\det(\lambda I - \mathsf{H}_m(L)A_m)}{\det \mathsf{H}_m(L)} = Q(\lambda)F(\lambda) - R(\lambda),$$

where

$$Q = \prod_{i=2}^{m-1}\left(1 + L\frac{K_{i-1}}{K_i}\lambda\right), \quad F = \lambda^3 + 11\lambda^2 + 31\lambda + 16, \quad R = 5\lambda^2 + 33\lambda + 16.$$

For $x \ge 0$, every factor in Q has modulus at least $|1 + L\lambda|$, because, with $\theta_i = K_{i-1}/K_i > 1$,

$$|1 + L\theta_i\lambda|^2 - |1 + L\lambda|^2 = 2L(\theta_i - 1)x + L^2(\theta_i^2 - 1)(x^2 + y^2) \ge 0.$$

If $\nu \ge 2$, then

$$|Q|^2 \ge |1 + L\lambda|^{2\nu} \ge 1 + 2\nu Lx + \nu L^2 y^2 \ge 1 + Ax + \frac{5}{4}y^2, \qquad A = 2\nu L.$$

On this domain $A \ge \sqrt{5\nu} \ge \sqrt{10} > 1/4$. The exact difference

$$\left(1 + Ax + \frac{5}{4}y^2\right)|F|^2 - |R|^2$$

has 22 nonzero monomials in x and y^2. After writing $A = U + 1/4$, every coefficient is nonnegative, and the difference is strict away from $\lambda = 0$. Thus no nonzero homogeneous eigenvalue lies in the closed right half-plane. Moreover $(QF - R)'(0) = 16L\sum_{i=2}^{m-1} K_{i-1}/K_i - 2 > 0$, so the conservation zero is simple. For $\nu = 1$, putting $\gamma = 91L/90$ gives

$$\frac{QF - R}{\lambda} = \gamma\lambda^3 + (1 + 11\gamma)\lambda^2 + (6 + 31\gamma)\lambda + (16\gamma - 2).$$

Here $\gamma > 1/8$ and the remaining cubic Routh–Hurwitz determinant is $6 + 99\gamma + 325\gamma^2 > 0$. Hence the primary mode and all higher-mode conclusions hold throughout the certified interval. The interval is nonempty: the case $\nu = 1$ is immediate, while for $\nu \geq 2$ one has $\sqrt{5/(4\nu)} < 4/5 < 90\nu/(90\nu + 1)$.

The proof then follows the physical-to-normalized chain explicitly. The positive equilibrium is $x^* = \mathsf{H}_m(L)^{-1}\mathbf{1}$, the physical diffusion is $D_m^{\mathrm{phys}} = \mathsf{H}_m(L)\Delta_m$, and the change $\widehat{x} = \mathsf{H}_m(L)x$ produces the row-scaled mode operator $\mathsf{H}_m(L)(A_m - t\Delta_m)$. The left critical vector becomes $\widetilde{\ell}(L) = \mathsf{H}_m(L)^{-1}\ell$ and the fixed-mass covector becomes $\mathsf{H}_m(L)^{-1}c$. The second harmonic is unchanged by the invertible row factor. The homogeneous correction changes by a kernel gauge:

$$w_0(L) = w_0^{\mathrm{ref}} + \tau_m(L)\rho_m, \qquad \tau_m(L) = -\frac{(\mathsf{H}_m(L)^{-1}c)^T w_0^{\mathrm{ref}}}{(\mathsf{H}_m(L)^{-1}c)^T \rho_m}. \tag{60}$$

The exact cubic numerator is

$$N_m(L) = N_m^{\mathrm{ref}} + \tau_m(L)S_m. \tag{61}$$

The symbolic bounds $N_m^{\mathrm{ref}} > 1/100$, $-1/10 < S_m < 0$, and $\tau_m(L) < 1/20$ imply $N_m(L) > 1/200$. Their complete shifted coefficient lists are printed in the Supplement and regenerated from the repository root by `python independent_verifier/verify_symbolic_certificates.py`. Meanwhile

$$\ell^T \mathsf{H}_m(L)^{-1} r = -\frac{485873}{924210} - \frac{11180}{1467} L\nu < 0.$$

At the first-mode crossing, invertibility of $\mathsf{H}_m(L)$ and (45) give

$$\ker[\mathsf{H}_m(L)(A_m - \Delta_m)] = \ker(A_m - \Delta_m) = \mathrm{span}\{r\}.$$

Its left nullvector is $\widetilde{\ell}(L)$. If a generalized eigenvector v satisfied $\mathsf{H}_m(L)(A_m - \Delta_m)v = r$, left multiplication by $\widetilde{\ell}(L)^T$ would give $0 = \widetilde{\ell}(L)^T r$, contradicting the displayed strict inequality. Thus the scaled critical zero is algebraically simple. The transformed transversality numerator is the unit-profile numerator:

$$\widetilde{\ell}(L)^T \mathsf{H}_m(L)\Delta_m r = \ell^T \Delta_m r < 0.$$

Consequently

$$\eta_m(L) = \frac{\widetilde{\ell}(L)^T \mathsf{H}_m(L)\Delta_m r}{\widetilde{\ell}(L)^T r} > 0,$$

and the scaled cubic coefficient is explicitly

$$c_m(L) = \frac{N_m(L)}{\widetilde{\ell}(L)^T r} < 0.$$

The branch and stability proof is the same sectorial exchange-of-stability argument as in Theorem 6.1. Diagonal concentration scaling does not alter the spatial reflection $\xi \mapsto \pi - \xi$,

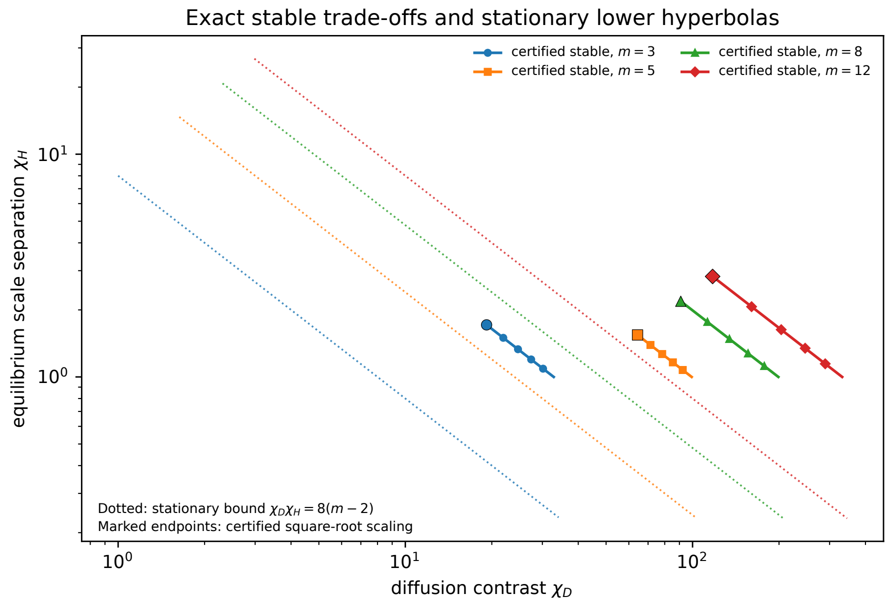


Figure 2: Exact heterogeneity trade-offs for representative dimensions. The dotted hyperbolas are the topology-specific necessary bounds for stationary crossings of $\widehat{N}_m$, $\chi_D\chi_H = 8(m-2)$; the solid curves are certified stable families with product $23(91(m-2)-1)/63$. Each marked point is the certified square-root-scaling endpoint for its dimension. Constants are not claimed optimal.

so the scaled reduced vector field is again odd and its two nonzero branches are exchanged by reflection. For each fixed m and L, the normalized branch converges to $\mathbf{1}$ in $H^2_N \hookrightarrow C^0$ and is componentwise strictly positive for sufficiently small $\mu > 0$. Multiplication by the positive diagonal matrix $\mathsf{H}_m(L)^{-1}$ preserves componentwise positivity in physical concentrations. The contrast formulas follow by comparing the explicit entries of $\mathsf{H}_m(L)$ and $D_m^{\mathrm{phys}}(L)$ on the interval (51). The first is strictly increasing and the second strictly decreasing in L. Moreover,

$$\frac{\chi_D(L_0)}{\chi_H(L_0)} = \begin{cases} 27209/2430, & \nu = 1, \\ \dfrac{136045\nu}{36(91\nu - 1)}, & \nu \geq 2, \end{cases} > 1.$$

Hence $\chi_D(L) > \chi_H(L)$ throughout the interval and the maximum is uniquely minimized at L_0. Finally (58) is the geometric-mean consequence of (35). $\square$

m	n	χ_D^{unit}	χ_D^{scale}	χ_H^{scale}	product	lower	η_m	c_m	$\sqrt{-\eta_m/c_m}$
3	4	32.8571	19.1809	1.71302	32.8571	2.82843	0.0192752	-0.00244362	2.80855
4	5	66.0794	52.5289	1.25796	66.0794	4	0.00992406	-0.00135495	2.70635
5	6	99.3016	64.3346	1.54352	99.3016	4.89898	0.00668208	-0.000932507	2.67689
6	7	132.524	74.2871	1.78394	132.524	5.65685	0.00503668	-0.000710298	2.66288
7	8	165.746	83.0556	1.9956	165.746	6.32456	0.00404149	-0.000573473	2.65469
8	9	198.968	90.9828	2.18688	198.968	6.9282	0.00337468	-0.000480799	2.64932
9	10	232.19	98.2727	2.36272	232.19	7.48331	0.00289675	-0.000413892	2.64553
10	11	265.413	105.058	2.52635	265.413	8	0.0025374	-0.000363322	2.6427

Table 1: Current-profile diagnostics generated from one exact source. χ_D^{unit} is the current unit-equilibrium stable-profile contrast; χ_D^{scale} and χ_H^{scale} are the physical contrasts at the certified square-root-scaling endpoint; "product" is their exact product $23(91(m-2)-1)/63$; "lower" is $\sqrt{8(m-2)}$; and η_m, c_m, and $\sqrt{-\eta_m/c_m}$ use the current unit-profile normal-form convention. The identity $\chi_D^{\text{unit}} = \chi_D^{\text{scale}}\chi_H^{\text{scale}}$ is exact, not a numerical coincidence. The table is illustrative and proves no all-dimensional claim.

8 The nonlinear frontier and finite-dimensional transparency

The exact unit-equilibrium stable infimum is open. The completed bounds are

$$8(m-2) \le \chi_{\text{stable}}(m) \le \frac{23(91m-183)}{63}. \tag{62}$$

A canonical near-threshold affine path, defined in Supplement Section S9, has, at $m = 3$,

$$c_3(\varepsilon) = \frac{6}{1379} + \frac{421985}{11409846}\varepsilon + O(\varepsilon^2), \qquad c_3(\varepsilon) > 0 \quad (0 < \varepsilon \le 10^{-3}). \tag{63}$$

Thus approaching the sharp linear threshold does not automatically preserve supercriticality. This is a boundary example, not a universal nonlinear gap.

Proposition 8.1 (Retuned local robustness)**.** *Fix m and a certified L. Sufficiently small perturbations within the positive-equilibrium realization manifold—equivalently, of positive steady fluxes and equilibrium coordinates with rates reconstructed accordingly—as well as small perturbations of diffusion ratios and interval length, admit a unique nearby scalar diffusion multiplier at which the primary supercritical bifurcation and local exponential stability persist.*

Proof. Continue the simple critical eigenvalue as a smooth function of the physical parameters and one scalar diffusion multiplier. Its multiplier derivative is the nonzero transversality pairing. The implicit function theorem gives the nearby critical multiplier and equilibrium. Homogeneous and finitely many low-mode gaps persist by analytic perturbation. In a common neighborhood, a positive diffusion lower bound and a reaction-Jacobian upper bound control all high modes. The reduced normal-form coefficients and branch spectral projection depend smoothly on parameters, so the negative cubic sign and complementary branch gap persist. One scalar multiplier is retuned; arbitrary nearby parameter points are not asserted to be critical. □

Open problems.

1. Determine whether $\chi_{\text{stable}}(m) = 8(m-2)$ as an infimum at unit equilibrium.
2. Determine whether supercritical stable pattern formation has a strict nonlinear contrast gap above the linear threshold.
3. Determine the constant-optimal stable diffusion–equilibrium frontier and, in particular, the exact infimum of $\max(\chi_D, \chi_H)$.
4. Determine the exact threshold for oscillatory diffusion-driven instability in this topology and whether wave instability can occur at lower diffusion or heterogeneity contrast than the sharp stationary threshold.

9 Numerical illustrations

The symbolic theorem does not use simulation. We integrated cosine–Galerkin systems for $m = 3, 5, 8$ near the exact unit-equilibrium branches, extracted the first-mode amplitude with the adjoint projection, and repeated the calculations under spatial and temporal refinement. Parameters, tolerances, profiles, and open-format data accompany the code.

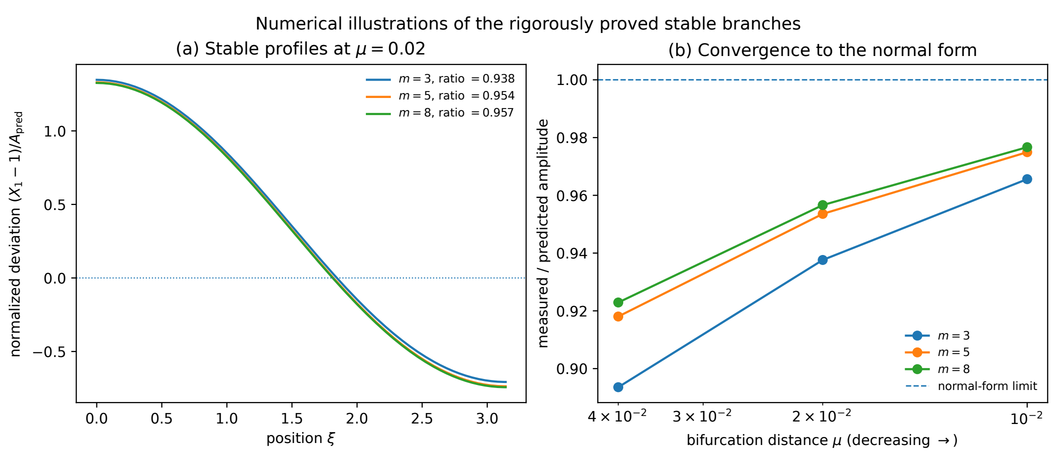


Figure 3: Numerical illustrations of the rigorously proved local patterned branches. Panel (b) compares the measured adjoint-projected amplitude with $\sqrt{-\eta_m \mu / c_m}$. Numerical illustrations are not used in any proof.

10 Relation to prior work and limitations

The possibility of order-$(n-1)$ unstable (activator) subsystems for general fixed matrices is established in multicomponent Turing theory [14]; that fixed-matrix possibility is not claimed as new. Anma, Sakamoto, and Yoneda give complementary three-component unstable-subsystem analysis [1]. The present distinction is the conjunction of one indexed binary-complex classical mass-action topology, every positive rate and equilibrium realization, all-spectrum stability of every smaller principal block, exact diffusion design, and stable nonlinear patterned branches in every dimension.

Haas and Goldstein quantified diffusion thresholds statistically for random Jacobians through six components and found that many sampled instabilities cannot be reduced to fewer diffusing species [5]. Here the scope is deterministic and complementary: one constrained mass-action topology is treated exactly in every dimension, with all positive realizations quantified and nonlinear branch stability proved.

Mincheva and Craciun's principal projected-network injectivity condition is a different reaction-network notion and is not identified with principal-block Hurwitz localization [11]. Interaction-topology and atlas studies identify broad structural features and finite collections of Turing-capable networks [10, 4, 15]; the present result instead gives one exact infinite topology family and a theorem uniform over all positive realizations. Marcon and collaborators showed that some architectures can pattern with equally diffusing signals under their modeling assumptions [10]. For the present topology, equal diagonal diffusion cannot destabilize: it only shifts the homogeneous spectrum left. Instead, the required stationary heterogeneity is quantified exactly by (31); at unit equilibrium its contrast floor is $8(m-2)$, and with equilibrium scaling the topology-specific stationary heterogeneity requirement is (35). This is a complementary, topology-specific quantitative result about differential-diffusion tuning, not a statement about all binary-complex networks. Parameter-rich unstable cores allow derivative freedoms absent from classical mass action [17]. Exact necessary-and-sufficient linear criteria for Turing and Turing–Hopf instabilities in general three-species systems address a different fixed-dimension scope [13]. Complementarily, Villar-Sepúlveda, Champneys, and Krause design Turing and wave instabilities of prescribed wavelength by choosing a generally non-diagonal diffusion tensor at fixed linearized kinetics, or conversely choosing linearized kinetics at fixed transport [20]; here diffusion is positive diagonal self-diffusion, and the Jacobians arise from—and are quantified over—all positive-equilibrium realizations of one indexed classical mass-action topology. Biochemical screens study interpretable reaction architectures rather than the synthetic extremal design here [12]. Conradi, Mincheva, and Uecker give sufficient spatial instability conditions for a restricted mass-action class and numerically continue nonlinear patterned branches, including numerically stable segments in a finite example [2]. Their work does not supply the present all-dimensional realization-wide localization theorem or exact contrast law.

The construction has one semipositive conservation law and an inflow; X_1 is not bounded by the conserved functional. The local stability theorem does not require an arbitrary-data global boundedness theorem. All diffusivities are positive and diagonal. We do not treat immobile species, cross-diffusion, projected-injectivity sharpness, weak reversibility, biochemical realization, far-from-onset dynamics, global attraction, or dimension-uniform robustness. The exact stable contrast constants remain open.

11 Discussion

The family is designed around two coupled positive-flux circuits. Their explicit two-parameter positive-flux factor $A_m(a,b)$, together with arbitrary positive right-diagonal equilibrium scaling H, makes the topology-wide quantifier tractable. The long chain forces every unstable principal subsystem to contain almost all species, while the boundary triad remains Hurwitz under arbitrary positive equilibrium scaling. The unique negative order-$(n-1)$ omission minor is the one omitting Z, which collapses stationary diffusion design to one coefficient. Finally, equilibrium scaling redistributes the necessary heterogeneity between concentrations and physical diffusion without changing the effective critical bracket or losing the negative cubic sign. Along the certified scaled family, this is a contrast allocation rather than a product reduction: $\chi_D(L)\chi_H(L) = \chi_D^{\mathrm{unit}}(m)$ is fixed. Within the certified interval, $\chi_D(L) > \chi_H(L)$, and their maximum is therefore minimized at the lower endpoint L_0, where it is $\Theta(\sqrt{m})$ rather than the unit-profile $\Theta(m)$.

The resulting conclusion is both structural and quantitative. The largest proper unstable-subsystem order is unavoidable within this mass-action class; the exact diffusion contrast is computable; stable positive patterns occur; and diffusion and equilibrium contrasts can be controlled simultaneously at the stationary-optimal square-root exponent for this topology. The remaining constant gap is explicit rather than hidden and provides a focused nonlinear design problem.

Data and code availability. Exact certificates, independent verifiers, mutation tests, numerical data, figure scripts, and replay instructions are frozen in the version 1.0.9 tagged release source tree. All archived versions are indexed by the stable Zenodo concept DOI; the exact preceding version 1.0.8 snapshot is archived at version DOI 10.5281/zenodo.22074358. The mutable development source remains on the main branch.

AI-assisted research disclosure. AI systems assisted with exploratory theorem discovery, symbolic derivation, code generation, adversarial testing, proof organization, and prose editing. AI systems are not authors. Central claims are supported by explicit proofs and independent exact implementations; neither numerical agreement nor AI confidence is treated as proof. The author assumes responsibility for all content.

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